
Research Profile	Scientific machine learning for magnetic inverse problems: probabilistic models, physical forward validation, and simulation-to-experiment robustness. Background in experimental condensed matter and computational many-body physics.	
Education	University of Illinois Urbana-Champaign	
	<i>Ph.D. in Physics, in progress</i>	Aug 2023–present
	Advisor: Bryan K. Clark	
	Xi'an Jiaotong University	
	<i>B.S. in Physics (Honors)</i>	Sep 2019–Jul 2023
	<i>Special Class for the Gifted Young</i>	Sep 2017–Jul 2019
	University of California, Berkeley	
	<i>BPIE Program (non-degree visiting program)</i>	Aug–Dec 2021
Current Research	University of Illinois Urbana-Champaign	
	<i>Graduate researcher, Bryan K. Clark group</i>	Nov 2025–present
	Crystal-field inverse modeling	
	• Represent non-unique crystal-field solutions from susceptibility with residual-MLP conditional Gaussian mixture models. • Validate candidates by reconstructing susceptibility; test training noise on synthetic curves and experimental measurements.	
	Magnetic-scattering inverse modeling	
	• Infer exchange parameters from powder-averaged neutron-scattering maps using MLP and ResNet conditional Gaussian mixtures. • Use exact Sunny forward checks and classical fits as references; test sensitivity to noise, intensity scale, and background offsets.	
	Neural-network backflow / variational Monte Carlo	
	• Implement JAX neural-network backflow wavefunctions and variational Monte Carlo for lattice Hubbard models. • Profile parallel GPU sampling, local-energy evaluation, and gradient synchronization; check chunked evaluation against the original computation.	

Publications	[P1] Evidence of intertwined pair density and charge density wave orders in UTe₂ Zhen Zhu, Yudi Huang, Julian May-Mann, Kaiming Liu, Zheyu Wu, Shanta R. Saha, Johnpierre Paglione, Alexander G. Eaton, Andrej Cabala, Michal Vališka, Eduardo Fradkin, Vidya Madhavan. <i>PNAS</i> 123 (29), e2602117123 (2026). arXiv.	
Prior Research	University of Illinois Urbana-Champaign <i>Research assistant, Vidya Madhavan group</i> Aug 2023–Oct 2025 • Grew thin-film heterostructures with MBE and characterized electronic structure using STM/STS in ultrahigh vacuum. • Automated growth temperature control and built Python/MATLAB analysis of STM/STS quasiparticle-interference data. Xi'an Jiaotong University <i>Research assistant, Yongchang Zhang group</i> Mar 2022–Jul 2023 • Simulated Rydberg-atom dynamics and photon-mediated interactions in Python/MATLAB to study quantum nonlinear optics. • Built a discrete-time lattice model of two-qubit phase-flip dynamics and many-body regimes. University of California, Berkeley <i>Research assistant, Michael F. Crommie group</i> Oct–Dec 2021 • Assisted synthesis and STM characterization of graphene nanoribbons with five-membered rings to study their electronic states.	
Teaching	University of Illinois Urbana-Champaign <i>Teaching assistant, PHYS 212</i> Spring & Fall 2026 <i>Teaching assistant, PHYS 102</i> Aug–Dec 2023	
Selected Honors	China National Scholarship 2021 Xi'an Jiaotong University Mathematical Contest in Modeling, Meritorious Winner 2021 COMAP	
Technical Skills	<i>ML / scientific computing</i>: Python (PyTorch, JAX, NumPy, SciPy), MATLAB; conditional mixture models; Git, Linux. <i>Computational physics / HPC</i>: MCMC, variational Monte Carlo, numerical optimization; GPU/cluster computation and runtime profiling. <i>Experiment</i>: MBE, STM/STS, ultrahigh vacuum, feedback control, experimental automation and data analysis.	